1. $g(x) = f(x) - 2v(x)$.

(A) 反例: $u(x)=2v(x)$ $f(x)=3v(x)$ $g(x)=v(x)$

$\lim\limits_{x\to x_0} f(x)$ 不存在, $\lim\limits_{x\to x_0} g(x)$不存在.

(B) 反例: $u(x)=v(x)$, $f(x)=2v(x)$ $g(x)=0$

$\lim\limits_{x\to x_0} f(x)$ 不存在, $\lim\limits_{x\to x_0} g(x)$ 存在.

(C) 正确.

(D) 反例: $u(x)=-v(x)$, $f(x)=0$ $g(x)=-2v(x)$.

$\lim\limits_{x\to x_0} f(x)$ 存在, $\lim\limits_{x\to x_0} g(x)$ 不存在.

2. $\lim\limits_{x\to 0} \dfrac{f''(x)}{|x|} = 1$ $\Rightarrow$ x 在 0 点的邻域内有 $f''(x)=|x|$

$f'(x) = \frac{1}{2}x^2 + C$ $f'(0)=0$ $f'(x)=\frac{1}{2}x^2 > 0$

$\therefore f(0)$ 是 $f(x)$ 的极小值点.

选 A

3. $\lim\limits_{x\to 0} f(x) = \lim\limits_{x\to 0} \int_0^x t^2\,dt = \lim\limits_{x\to 0} \frac{1}{3}x^3$

$f(x)$ 是 $g(x)$ 的低阶无穷小量.

选 D.

4. I_1：令 $f_1(x)=\dfrac{\sin x}{1+x^2}\cos^4 x$　　$f_1(-x)=\dfrac{-\sin x}{1+x^2}\cos^4 x=-f_1(x)$.

$I_1=0$.

$$I_2=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\underbrace{\frac{\sin^3 x}{\text{奇函数}}}\,dx+\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4 x\,dx=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4 x\,dx.$$

$$I_3=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}x^2\sin^3 x\,dx-\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4 x\,dx=-\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4 x\,dx.$$

$\cos^4 x\geqslant 0$　　$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^4 x\,dx>0$

$\therefore$　$I_3<I_1<I_2$.　选　D

5. (A)　×　　$\lim\limits_{x\to a}f(x)$ 可能不存在　如 $f(x)=\frac{1}{x}$　$(0,1)$

(B).　×　　极值条件与 $f''(x_0)$ 无关，反例：$f(x)=x^4$.

(C)　√　　$\vec{a}\cdot\vec{b}=0\Rightarrow\vec{a}\perp\vec{b}$　$\vec{a}\times\vec{c}=0\Rightarrow\vec{a}\parallel\vec{c}$.

$\therefore\vec{b}\perp\vec{c}$　$\therefore\vec{b}\cdot\vec{c}=0$

(D)　×　　令　$F(x)=\int_0^x f(t)\,dt$

$$\int_0^{g(x)}f(t)\,dt=F(g(x))$$

$$\frac{d\left(\int_0^{g(x)}f(t)\,dt\right)}{dx}=\frac{d(F(g(x))}{dx}=(F(g(x)))'$$

$$=F'(g(x))\cdot g'(x)$$

$$=f(g(x))\cdot g'(x).$$

二. 1. $\lim\limits_{x\to 0+} (e^{2x}-1)^{\frac{1}{\ln x}} = \lim\limits_{x\to 0+} e^{\frac{1}{\ln x}\cdot \ln(e^{2x}-1)}$

$= e^{\lim\limits_{x\to 0+} \frac{\ln(e^{2x}-1)}{\ln x}}$

$\lim\limits_{x\to 0+} \frac{\ln(e^{2x}-1)}{\ln x} = \lim\limits_{x\to 0+} \frac{\frac{2e^{2x}}{e^{2x}-1}}{\frac{1}{x}} = \lim\limits_{x\to 0+} \frac{2x\cdot e^{2x}}{e^{2x}-1}$

$= \lim\limits_{x\to 0+} \frac{2e^{2x}+4x\cdot e^{x}}{2e^{2x}} = 1$

原式 $= e$

2. $$x_n = \sum_{i=1}^{n} \frac{i}{3n^2+i} = \sum_{i=1}^{n} \frac{\frac{i}{n}}{3n+\frac{i}{n}} = \int_0^1 \frac{x}{3n+x}\,dx$$

$$= \int_0^1 1 - \frac{3n}{3n+x}\,dx = \int_0^1 1\,dx - 3n\int_0^1 \frac{1}{3n+x}\,d(3n+x).$$

$$= x\Big|_0^1 - 3n\cdot\left(\ln|3n+x|\Big|_0^1\right).$$

$$= 1 - 3n\cdot\ln\frac{3n+1}{3n}$$

3. $$y' = f'\cdot(1+y')$$

$$y' = \frac{f'}{1-f'}$$

$$y'' = \frac{f''(1+y')(1-f') + f''(1+y')\cdot f'}{(1-f')^2}$$

$$= \frac{f''(1+y')}{(1-f')^2} = \frac{f''\left(\frac{f'}{1-f'}+1\right)}{(1-f')^2} = \frac{f''}{(1-f')^3}$$

$$\therefore \frac{d^2y}{dx^2} = \frac{f''(x+y)}{(1-f'(x+y))^3}$$

4. $(1+\frac{1}{n})^n = e^{n\ln(1+\frac{1}{n})}$

$\ln(1+\frac{1}{n}) = \frac{1}{n} - \frac{1}{2n^2} + o(\frac{1}{n^2})$.

$(1+\frac{1}{n})^n = e^{n\cdot(\frac{1}{n} - \frac{1}{2n^2} + o(\frac{1}{n^2}))} = e^{1-\frac{1}{2n}+o(\frac{1}{n})}$

$\lim_{n\to\infty} n\left(1 - \frac{(1+\frac{1}{n})^n}{e}\right) = \lim_{n\to\infty} n\left(1 - \frac{e^{1-\frac{1}{2n}+o(\frac{1}{n})}}{e}\right)$.

$= \lim_{n\to\infty} n\left(1 - e^{-\frac{1}{2n}+o(\frac{1}{n})}\right)$

$= \lim_{n\to\infty} n\left(1 - (1 - \frac{1}{2n} + o(\frac{1}{n}))\right)$

$= \frac{1}{2}$

5. $\int \frac{dx}{x+\sqrt{a^2-x^2}}$ 令 $x = a\sin t$

$= \int \frac{a\cos t\ dt}{a\sin t + a\cos t}$

$= \int \frac{\cos t\ dt}{\sin t + \cos t}$

$= \int \frac{\sin t}{\sin t+\cos t}\ dt$

$= \frac{1}{2}\int \frac{\cos t}{\sin t+\cos t} + \frac{\sin t}{\sin t+\cos t}\ dt$

$= \frac{1}{2}\int 1\ dt$

$= \frac{t}{2} + C$

$= \frac{\arcsin x}{2a} + C$

6. $x'(t) = -a\sin t$

$y'(t) = a\cos t$

$z'(t) = b$.

弧长. $S = \int_0^{2\pi} \sqrt{a^2\sin^2 t + a^2\cos^2 t + b^2}\, dt$

$= \int_0^{2\pi} \sqrt{a^2+b^2}\, dt$

$= \sqrt{a^2+b^2} \cdot 2\pi$.

切线: 点 $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}, \frac{\pi b}{4}\right)$: $t = \frac{\pi}{4}$.

$x'\left(\frac{\pi}{4}\right) = -\frac{a}{\sqrt{2}}$ $y'\left(\frac{\pi}{4}\right) = \frac{a}{\sqrt{2}}$ $z'\left(\frac{\pi}{4}\right) = b$

切线方程: $\frac{x - \frac{a}{\sqrt{2}}}{-\frac{a}{\sqrt{2}}} = \frac{y - \frac{a}{\sqrt{2}}}{\frac{a}{\sqrt{2}}} = \frac{z - \frac{\pi b}{4}}{b}$.

法平面: $-\frac{a}{\sqrt{2}}\left(x - \frac{a}{\sqrt{2}}\right) + \frac{a}{\sqrt{2}}\left(y - \frac{a}{\sqrt{2}}\right) + b\left(z - \frac{\pi b}{4}\right) = 0$.

7. 过面 $x+5y+z=0$ 法向量 $(1,5,1)$

过面 $x-z+4=0$ 法向量 $(1,0,-1)$

设目标平面法向量为 ~~[illegible]~~ $n=(1+k,5,1-k)$

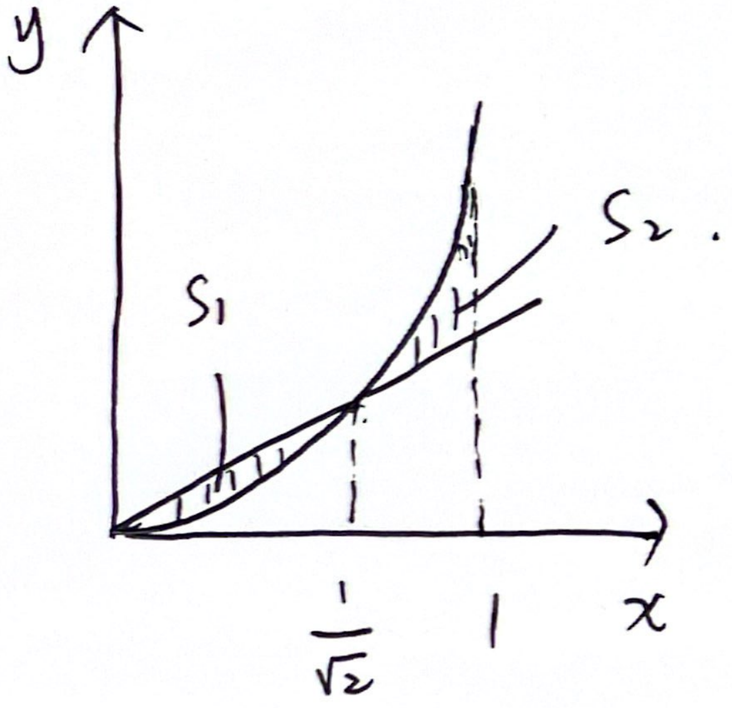
面 $x-4y-8z+12=0$ 法向量 $m=(1,-4,-8)$

$$\frac{|n\cdot m|}{|n|\cdot|m|}=\frac{\sqrt{2}}{2}=\frac{|9k-27|}{\sqrt{2k^2+27}\cdot 9}=\frac{\sqrt{2}}{2}\qquad k=-\frac{3}{4}$$

平面: $\frac{1}{4}x+5y+\frac{7}{4}z+d=0 \Rightarrow d=-3$ $\frac{1}{4}x+5y+\frac{7}{4}z-3=0$.

交点 $\frac{x}{\sqrt{2}} = x^2$ $x=0$ 或 $\frac{1}{\sqrt{2}}$

8.



$$V_1 = \int_0^{\frac{1}{\sqrt{2}}} \frac{1}{2}x^2\,dx - \int_0^{\frac{1}{\sqrt{2}}} x^4\,dx$$

$$V_2 = \int_{\frac{1}{\sqrt{2}}}^{1} x^4\,dx - \int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{2}x^2\,dx$$

$$V = V_1 + V_2 = \frac{\sqrt{2}+1}{30}\pi$$

三.

1. $x \neq 0$

$$y' = e^{\frac{1}{x}} + e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right) \cdot (x+6)$$

$$= e^{\frac{1}{x}} \left(1 - \frac{1}{x} - \frac{6}{x^2}\right)$$

$$= -e^{\frac{1}{x}} \left(\frac{2}{x} + 1\right)\left(\frac{3}{x} - 1\right)$$

$x < -2$ $\quad y' > 0$ ↗

$-2 < x < 0$ $\quad y' < 0$ ↘. $\quad 0 < x < 3$ $\quad y' < 0$ ↘.

$x > 3$ $\quad y' > 0$ ↗

极值: $x = -2$ $\quad 4e^{-\frac{1}{2}}$

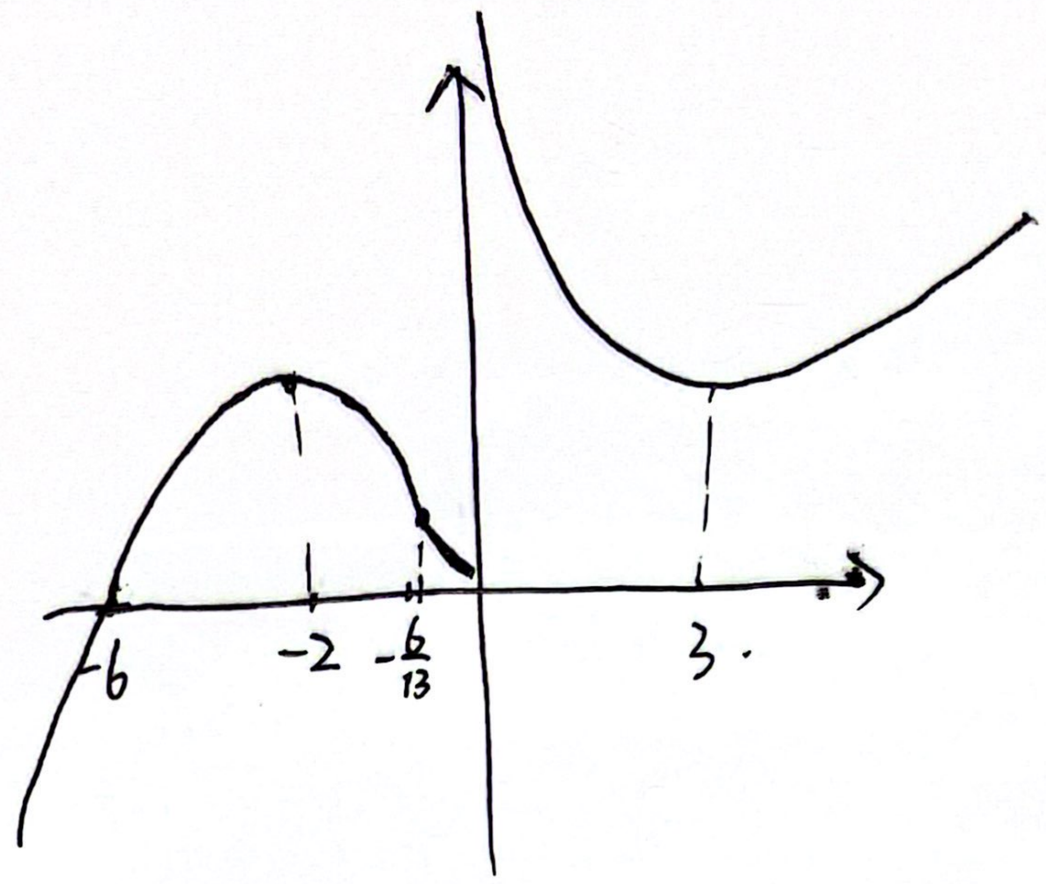
$x = 3$ $\quad 9e^{\frac{1}{3}}$

2. $y'' = \dfrac{13x+6}{x^4} e^{\frac{1}{x}}$

$x < -\frac{6}{13}$ $\quad y'' < 0$ $\quad$ 上凸

$y > -\frac{6}{13}$ $\quad y'' > 0$. $\quad$ 下凸.

3.



$$\lim_{x\to 0+} f(x) = +\infty$$

有渐近线 $x=0$.

$$\lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{x+6}{x} e^{\frac{1}{x}} = 1$$

$$\lim_{x\to\infty} f(x)-x = \lim_{x\to\infty} (x+6)e^{\frac{1}{x}} - x$$

$$= \lim_{x\to\infty} x e^{\frac{1}{x}}\left(1+\frac{6}{x}-e^{-\frac{1}{x}}\right)$$

$$= \lim_{x\to\infty} e^{\frac{1}{x}}\left(x+6-xe^{-\frac{1}{x}}\right).$$

$t=\frac{1}{x}$

$$= \lim_{t\to 0} e^{t}\left(6+\frac{1-e^{-t}}{t}\right)$$

$$= \lim_{t\to 0} e^{t}\left(6+\frac{e^{t}}{1}\right)$$

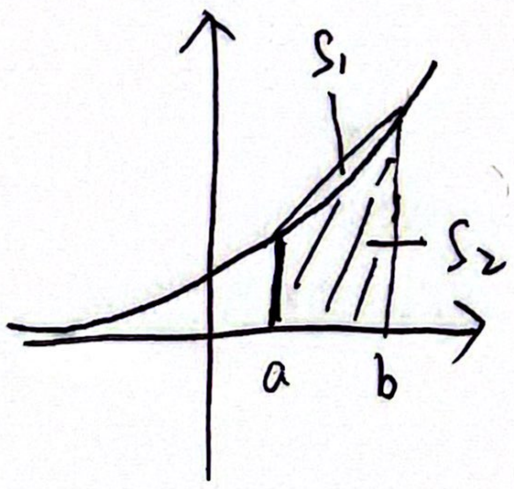
$$= 7$$

有渐近线 $y=x+7$

$x=0$.

四. 不妨设 $a>b$. $f(x)=e^x$

$$\frac{e^a-e^b}{a-b} < \frac{e^a+e^b}{2} \iff e^a-e^b < \frac{a-b}{2}(e^a+e^b)$$



$$e^a-e^b = \int_b^a e^x\,dx = S_2$$

$$\frac{a-b}{2}(e^a+e^b) = S_1+S_2$$

$$\text{事可以证明, } g(x) = \frac{e^a-e^b}{a-b}x + \frac{ae^b-be^a}{a-b} \geq f(x).$$

在 $[a,b]$ 上, $h(x)=g(x)-f(x)$

$$h'(x) = \frac{e^a-e^b}{a-b} - e^x$$

微分中值: $\exists\ c\in[a,b]$ $e^c = \frac{e^a-e^b}{a-b}$

$x<c$ $h'(x)<0$ ↓ $x>c$ $h'(x)>0$

$h(a)=h(b)=0$ $\therefore h(x)\leq 0$ $x\in[a,b]$

五. 1. $\because \lim\limits_{x\to a^+} \dfrac{f(2x-a)}{x-a}$ 存在

$\lim\limits_{x\to a^+} f(2x-a) = 0 \quad \therefore f(a)=0$

$f'(x)>0 \quad f(x)$ 在 (a,b) 上 递增.

$\therefore f(x)>0$ 在 (a,b) 上

2. 令 $g(x)=x^2 \quad F(x)=\int_a^x f(t)\,dt. \quad F'(x)=f(x).$

$F(b)-F(a)=\int_a^b f(x)\,dx.$

柯西中值: $\dfrac{b^2-a^2}{\int_a^b f(x)dx} = \dfrac{g(b)-g(a)}{F(b)-F(a)} = \dfrac{g'(c)}{F'(c)} = \dfrac{2c}{f(c)}$

3. $f(\xi)=f(\xi)-0=f(\xi)-f(a)$

微分中值: $\exists\, \eta\in(a,\xi) \quad f(\xi)=f'(\eta)(\xi-a).$

由2. $\dfrac{b^2-a^2}{\int_a^b f(x)dx} = \dfrac{2\xi}{f(\xi)} = \dfrac{2\xi}{f'(\eta)(\xi-a)}$

$\therefore f'(\eta)(b^2-a^2) = \dfrac{2\xi}{\xi-a}\int_a^b f(x)dx$